

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110006

Date: 05-01-2019

Subject Name: Basic Mechanical Engineering

Time: 10:30 am to 01:00 pm

Total Marks: 70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Discuss Closed, Open and Isolated Thermodynamic system with neat sketch.	03
	(b) What is solid fuel? Discuss different types of solid fuel.	04
	(c) Describe Isothermal process and derive expression for Workdone, Change in Internal energy, Heat transfer and Change in Enthalpy.	07
Q.2	(a) Dryness fraction of steam cannot have the value more than unity: Justify	03
	(b) Explain Steam formation T-h plot	04
	(c) The heat transfer from a heat reservoir is proportional to its temperature: Justify by deriving equation.	07
	OR	
	(c) 1.5kg of steam at a pressure of 10bar and temperature of 250°C is expanded until the pressure becomes 2.8bar. The dryness fraction of steam is then 0.9. Calculate change in Internal Energy	07
Q.3	(a) List different mountings of boiler and explain any one in brief.	03
	(b) Explain: Smoke tube internally fired horizontal type stationary boiler.	04
	(c) The efficiency of Otto cycle is a function of compression ratio: Prove it.	07
	OR	
Q.3	(a) What is throttling calorimeter? Explain its limitation.	03
	(b) Discuss Rankine cycle with block diagram	04
	(c) What is split AC? How it works? Explain with advantage and disadvantage.	07
Q.4	(a) Explain the term: (i) Swept volume (ii) Clearance volume (iii) Stroke length	03
	(b) Discuss with neat sketch Diaphragm pump.	04
	(c) Distinguish between Reciprocating and Rotary Compressor.	07
	OR	
Q.4	(a) Explain with neat sketch single acting Plunger type pump.	03
	(b) Explain need of multi staging in reciprocating air compressor with its advantages.	04

- (c) The following readings were taken during the test on a single cylinder four stroke IC engine: **07**
- | | |
|----------------------------------|--------------|
| Cylinder diameter | : 270mm |
| Stroke Length | : 380mm |
| Mean Effective Pressure | : 6bar |
| Engine speed | : 250rpm |
| Net load on brake | : 1000N |
| Effective mean diameter of brake | : 1.5m |
| Fuel used | : 10kg/hr |
| Calorific value of Fuel | : 44400kJ/kg |
- Calculate:
- Brake Power
 - Indicated Power
 - Mechanical Efficiency
 - Indicated thermal efficiency

- Q.5 (a) Discuss the term: **03**
- Condenser
 - Baffle tray
 - Evaporator
- (b) Explain with neat sketch the working of Internal Expanding Shoe Brake. **04**
- (c) Discuss the following with application and Properties: **07**
- Glass
 - Ceramic
 - Plastics

OR

- Q.5 (a) How metals are classified? Show with block diagram **03**
- (b) Give brief comparison between Belt, Chain and Gear drive. **04**
- (c) Explain with sketch: **07**
- Centrifugal clutch
 - Fast and Loose pulley drive

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110006****Date: 04/06/2019****Subject Name: Basic Mechanical Engineering****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of steam table is permitted.

		Marks
Q.1	(a) Define Melting point, Boiling point and Trippl point of water using p-v diagram.	03
	(b) Derive an expression for internal energy for a closed system.	04
	(c) Define specific heat at Constant volume, constant pressure and Adiabatic index. Also derive relationship between specific heats in form of Characteristic gas constant.	07
Q.2	(a) Discuss the factors responsible for global warming and Ozone depletion	03
	(b) Explain Equivalent evaporation and factor of evaporation	04
	(c) Determine the value of final dryness fraction of steam 1. After losing 125 kJ from the steam at constant pressure. 2. After expansion to 3 bar pressure in a turbine stage and work equivalent of 20 kJ/kg done Initially steam is available at 7 bar pressure and 0.9 dryness fraction.	07
	OR	
	(c) Three kg of steam at a pressure of 10 bar exists in the following conditions. Calculate its enthalpy and internal energy in each of the cases. 1. Steam with $x = 0.91$ 2. Steam at temperature 200°C	07
Q.3	(a) Explain the construction and function of Steam Trap with neat sketch. Also mention its specific location in the system.	03
	(b) Derive the equation for efficiency of Carnot	04
	(c) A double acting reciprocating pump has piston diameter of 150 mm and stroke length of 225 mm. the suction and delivery heads are 4 m and 12 m respectively. If the speed of the pump is 80 rpm and the actual quantity of water discharged is $0.61\text{ m}^3/\text{min}$. find the percentage slip, the coefficient of discharge and the power required to drive the pump if the efficiency of the pump is 80%.	07
	OR	
Q.3	(a) Explain the construction and function of Steam separator with neat sketch. Also mention its specific location in the system.	03
	(b) Prove that the efficiency of Otto cycle is greater than that of Diesel cycle for the same compression ratio.	04
	(c) Air is to be compressed in a single stage reciprocating compressor from 1.013 bar and 15°C to 7 bar. Calculate the indicated power required for free air delivery of $0.3\text{ m}^3/\text{min}$ when the compression process is 1. Isentropic 2. Reversible isothermal 3. Polytropic with $n = 1.25$.	07

- Q.4** (a) Draw a neat sketch of p-v diagram for single stage compressor with clearance. **03**
- (b) It is required to produce ice from 36° water. The capacity of the ice plant is 6 ton and specific heat of the water is 4.18 KJ/kg.K. Determine the amount of ice produced in 3 hours. **04**
- (c) In an ideal Otto cycle the air at the beginning of isentropic compression is at 1 bar and 15°C. The ratio of compression is 8. If the heat added during the constant volume process is 1000 kJ/kg, determine (a) the maximum temperature in the cycle, (b) the air standard efficiency (C) work done per kg of air. **07**

OR

- Q.4** (a) Draw a neat sketch of p-v diagram showing Free Air Delivery for air compressor. **03**
- (b) Calculate the energy consumed in one month for following conditions: **04**
 COP of air-conditioning unit : 5
 Capacity of air conditioner : 2 TR
 No of air conditioners : 8
 All air conditioners run for 4 hours/day.
- (c) A petrol engine with a stroke length of 200 mm and diameter of 150 mm has a clearance volume of $7 \times 10^5 \text{ mm}^3$. If the indicated thermal efficiency is 0.30, find the relation efficiency. If the effective pressure is 5 bar and engine runs at 1000 rpm. Find the IP of the engine. take $\gamma=1.4$ **07**
- Q.5** (a) Discuss the operational difference between Vapour compression and Vapour absorption refrigeration cycle **03**
- (b) Define velocity ratio of pulleys and discuss effect of slip and creep on motion transmission, **04**
- (c) Using neat sketch explain the working of Cone clutch and Centrifugal clutch **07**

OR

- Q.5** (a) Define following material properties. **03**
 1. Ductility
 2. Plasticity
 3. Malleability
- (b) Define pitch circle, diametral pitch, addendum and dedendum using neat sketch. **04**
- (c) Using neat sketch explain the working of Block brake and Internal expanding shoe brake. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110006****Date: 03/01/2020****Subject Name: Basic Mechanical Engineering****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of steam tables is allowable

		Marks
Q.1	(a) Define prime movers. Write any Four examples of the prime movers.	03
	(b) Define the terms: Hardness, Toughness, Ductility, Elasticity	04
	(c) List the non-convectional sources of energy. Prepare a short note on solar energy.	07
Q.2	(a) Distinguish between coupling and clutch.	03
	(b) Describe briefly using neat diagrams the types of belt drives.	04
	(c) Explain with neat sketch the Vapor Compression Refrigeration system. Also draw p-h and T-s diagram for the same.	07
	OR	
	(c) Classify Air Compressors. Explain construction and working of centrifugal pump with sketch.	07
Q.3	(a) Prove that the difference between two specific heats of gases is equal to its characteristic gas constant.	03
	(b) Differentiate between gas constant and Universal gas constant.	04
	(c) One kg of gas at a pressure of 100 kPa and temperature of 17 °C is compressed isothermally in a piston-cylinder arrangement to final pressure of 2500 kPa. The characteristic gas equation is given by the relation $pV = 260 T$ per kg where T is in Kelvin. Find out (1) Final Volume (2) Compression Ratio (3) Change in enthalpy (4) Work done on the gas.	07
	OR	
Q.3	(a) One kg gas is compressed adiabatically by following the law $pV^\gamma = C$ from initial temperature of 290K. The initial pressure of gas is 1 bar. The initial and final volumes of gas are 0.50 m ³ and 0.05 m ³ respectively. Find the final temperature and pressure of gas. Assume $\gamma=1.4$	03
	(b) Write the uses of “Steam Tables”	04
	(c) Find the internal energy of 1 kg of steam at a pressure of 15 bar when steam is (1) Super-heated at a temperature of 400 °C and (2) Wet with dryness fraction of 0.9. Take specific heat of superheated steam as 2.1 kJ/kg-K.	07
Q.4	(a) A diesel engine works on Diesel cycle with a compression ratio of 15 and cut off ratio of 1.75. Calculate the air standard efficiency. Assume $\gamma= 1.4$.	03
	(b) Derive the equation for air standard efficiency of Otto cycle.	04
	(c) The following readings were observed during a test on Two Stroke Single Cylinder Diesel engine:	07
	Bore	= 22 cm
	Stroke	= 28 cm
	Speed	= 350 rpm

Net brake load	=	65 kg
Effective brake drum diameter	=	100 cm
Mean Effective Pressure	=	3 bar
Fuel consumption	=	4 kg/h
Calorific value of fuel	=	43 MJ

Calculate: (1) Indicated Power (2) Brake Power (3) Mechanical Efficiency (4) Brake thermal Efficiency

OR

- Q.4** (a) Classify I.C. Engines. **03**
 (b) Determine the brake thermal efficiency of an engine with following data: **04**
 Brake Power = 80 kW
 Fuel consumption rate = 20 kg/hour
 Calorific value of the fuel = 43 MJ/kg
 (c) Discuss the construction and working of four stroke Petrol engines. **07**
- Q.5** (a) Define the terms 'Refrigeration', "Ton of Refrigeration" and "Coefficient of Performance" **03**
 (b) What is boiler? Compare fire tube boiler with water tube boiler. **04**
 (c) Explain with neat sketch, the construction and working of **07**
 (1) Fusible plug (2) Air preheater.

OR

- Q.5** (a) Differentiate between reciprocating compressor and rotary compressor. **03**
 (b) Differentiate between boiler mountings and accessories. **04**
 (c) Discuss the construction details and working of Cochran boiler with neat sketch. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE – SEMESTER 1&2 EXAMINATION – SUMMER 2020****Subject Code: 3110006****Date: 06/11/2020****Subject Name: Basic Mechanical Engineering****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		MARKS
Q.1	(a) Discuss Closed, Open and Isolated Thermodynamic system with neat sketch.	03
	(b) Define the terms: Hardness, Toughness, Ductility, Elasticity	04
	(c) Derive an expression for internal energy for a closed system	07
Q.2	(a) Distinguish between coupling and clutch.	03
	(b) Explain Equivalent evaporation and factor of evaporation	04
	(c) The heat transfer from a heat reservoir is proportional to its temperature: Justify by deriving equation.	07
	OR	
	(c) Explain construction and working of centrifugal pump with a neat sketch	07
Q.3	(a) Explain the construction and function of Steam Trap with neat sketch	03
	(b) Write the uses of “Steam Tables”	04
	(c) 1.5kg of steam at a pressure of 10bar and temperature of 250°C is expanded until the pressure becomes 2.8 bars. The dryness fraction of steam is then 0.9. Calculate change in Internal Energy	07
	OR	
Q.3	(a) List different mountings of boiler and explain any one in brief.	03
	(b) Derive the equation for air standard efficiency of Otto cycle.	04
	(c) A petrol engine with a stroke length of 200 mm and diameter of 150 mm has a clearance volume of $7 \times 10^5 \text{ mm}^3$. If the indicated thermal efficiency is 0.30, find the relation efficiency. If the effective pressure is 5 bar and engine runs at 1000 rpm. Find the IP of the engine. take $\gamma=1.4$	07
Q.4	(a) Discuss with neat sketch Diaphragm pump.	03
	(b) Distinguish between Reciprocating and Rotary Compressor.	04
	(c) Discuss the construction and working of four stroke Petrol engine.	07
	OR	
Q.4	(a) Draw a neat sketch of p-v diagram showing Free Air Delivery for air compressor.	03
	(b) Define the terms ‘Refrigeration’, “Ton of Refrigeration” and “Coefficient of Performance”	04

- (c) Calculate the energy consumed in one month for following conditions: COP of air-conditioning unit : 5
Capacity of air conditioner :
2 TR No of air conditioners : 8
All air conditioners run for 4 hours/day. **07**
- Q.5** (a) What is throttling calorimeter? Explain its limitation **03**
(b) Discuss with neat sketch any two boiler accessories. **04**
(c) Give comparison between Belt, Chain and Gear drive. **07**
- OR**
- Q.5** (a) How metals are classified? Show with block diagram **03**
(b) Define velocity ratio of pulleys and discuss effect of slip and creep on motion transmission **04**
(c) Discuss the following with application and Properties: **07**
 (i) Glass
 (ii) Ceramic
 (iii) Plastics

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II (NEW) EXAMINATION – WINTER 2020****Subject Code:3110006****Date:18/03/2021****Subject Name:Basic Mechanical Engineering****Time:10:30 AM TO 12:30 PM****Total Marks:56****Instructions:**

1. Attempt any FOUR questions out of EIGHT questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of steam tables is permitted in exam

		Marks
Q.1	(a) List advantages of Solar energy and its application.	03
	(b) Explain phenomena of global warming and causes of ozone depletion.	04
	(c) Define the following terms: Prime movers, Internal energy, heat capacity, first law of thermodynamics, state, Path, Zeroth law of thermodynamics.	07
Q.2	(a) Classify pumps on basis of principle, construction, and fluid flow direction in pump.	03
	(b) Explain the following terms: volumetric efficiency, compression ratio.	04
	(c) With usual notations derive expression to determine efficiency of Otto Cycle.	07
Q.3	(a) Define the following terms: Boyles law, Avogadro's law, Charles Law.	03
	(b) A Vessel of 2.5 m ³ capacity contains one kg –mole of nitrogen at 100°C. Calculate the specific volume and pressure if the gas is cooled to 30°C, calculate final pressure, change in specific internal energy and specific enthalpy.	04
	(c) Explain adiabatic process. Derive an expression for work done during the adiabatic expansion of an ideal gas.	07
Q.4	(a) What is superheated steam, list its advantages and application.	03
	(b) With neat sketch explain construction and working of separating calorimeter.	04
	(c) Calculate the internal energy of 1 kg of superheated steam at a pressure of 10 bar and 280°C. If this steam is to be expanded to a pressure of 1.6 bar and 0.8 kg dry, determine the change in internal energy. Assume specific heat of superheated steam as 2.1 k J /kg K.	07
Q.5	(a) In an Otto cycle the temperature at the beginning and end of the isentropic compression are 316 K and 596 K resply. Determine the air standard efficiency and compression ratio. Assume $\gamma=1.4$	03
	(b) With neat sketch explain construction and working of water level indicator.	04
	(c) Draw neat sketch of an internal combustion engine, list its parts and state functions of each.	07
Q.6	(a) Compare reciprocating compressor and rotary compressor.	03
	(b) List essential qualities of a good boiler.	04

- (c) During a trial on single cylinder oil engine, working on the four stroke cycle and fitted with rope brake the following readings were taken, Effective diameter of brake wheel= 630mm, dead load on the brake=200N, spring balance reading=30N, Speed =450rpm, Area of indicator diagram=420mm², length of indicator diagram=60mm, spring scale=1.1 bar per mm, diameter of cylinder=100mm, stroke=150mm, quantity of oil used =0.815kg/hr, calorific fuel of oil =42,000k J /kg, Calculate brake power, indicated power, mechanical efficiency, brake thermal efficiency, and brake specific fuel consumption. **07**
- Q.7** (a) Draw neat sketch of domestic refrigerator and list its different parts. **03**
 (b) Compare belt drive, chain drive and gear drive. **04**
 (c) List and explain physical, thermal, mechanical and electrical properties of metals. **07**
- Q.8** (a) Explain working of split air conditioner. **03**
 (b) List properties of copper and its applications. **04**
 (c) Draw sketch and label the following: shoe brake, cone clutch, flange coupling **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110006****Date:30/03/2022****Subject Name:Basic Mechanical Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q-1**
- (a) Define zeroth law of thermodynamics, and First law of thermodynamics. **03**
- (b) With usual notations prove that $C_p - C_v = R$. **04**
- (c) Write a difference between SI engine and CI engine. **07**

- Q-2**
- (a) Describe the process of formation steam on T-H diagram. **03**
- (b) Write a short note on “Global Warming and solar energy” **04**
- (c) Ford car having a four cylinder, four stroke petrol engine has 100 mm bore and stroke is 1.25 times the bore. It consumes 4 kg of fuel per hour having calorific value of 41,000 kJ/kg. The engine speed is 800 rpm. Calculate indicated thermal efficiency if mean effective pressure is 0.75 MPa. **07**

OR

- (c) A 4-cylinder, two-stroke cycle petrol engine develops 30 kW at 2500 rpm. The mean effective pressure on each piston is 8 bar and mechanical efficiency is 80%. Calculate the diameter and stroke of each cylinder if stroke to bore ratio is 1.5. Also the fuel consumption of engine, if brake thermal efficiency is 28% and calorific value is 43900 kJ/kg. **07**

- Q-3**
- (a) How do you classify steam boilers? **03**
- (b) Draw labeled diagram of Babcock and wilcox boiler. **04**
- (c) An air standard Otto cycle has compression ratio of 6. The temperature at the start of compression is 25°C and pressure is 1 bar. If the maximum temperature of the cycle is 1150°C. Calculate (a) the heat supplied and heat rejected per kg of air (b) network done per kg of air and (c) thermal efficiency of cycle. Assume $\gamma=1.4$, $C_v= 0.778$ kJ/kg K for air. **07**

OR

- Q-3**
- (a) List out application of compressed air. **03**
- (b) Explain single acting reciprocating pump. **04**
- (c) One kg of a gas at 1 bar pressure and 17°C is compressed isothermally to a pressure of 25 bar in a cylinder. The characteristic gas constant is 260 J/kg K. calculate (a) the final temperature (b) work done and (c) change in enthalpy. **07**

- Q-4**
- (a) What is priming? Why priming is required in centrifugal pump but not in reciprocating pump. **03**
- (b) Define (i) Dryness fraction (ii) wetness fraction. **04**
- (c) Explain vapour compression refrigeration cycle used in domestic refrigerator. **07**

OR

- Q-4**
- (a) Write function of clutch, Break and Coupling. **03**
- (b) Compare belt drive, chain drive and gear drive. **04**

- (c) Define following mechanical properties: (1) Elasticity (2) Malleability (3) Ductility (4) Stiffness (5) Hardness (6) Toughness (7) Resilience. **07**
- Q-5** (a) Explain open system, closed system and isolated system. **03**
(b) Define (i) one ton of refrigeration (ii) COP. **04**
(c) Explain construction and working of centrifugal compressor with neat sketch. **07**
- OR**
- Q-5** (a) Define ferrous and Nonferrous material with example. **03**
(b) Explain types of Belt Drives. **04**
(c) Write a short on a single plate (disc) friction clutch. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022****Subject Code:3110006****Date:03-08-2022****Subject Name:Basic Mechanical Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1**
- | | | |
|-----|---|-----------|
| (a) | Define (1) Critical Point (2) Enthalpy (3) Extensive property. | 03 |
| (b) | Give advantages of liquid fuels compared to solid fuels. | 04 |
| (c) | Derive the characteristics gas equation for a perfect gas with usual notations. | 07 |

- Q.2**
- | | | |
|-----|---|-----------|
| (a) | Give comparison between work and heat. | 03 |
| (b) | Define (1) sensible heat (2) Latent heat (3) Degree of superheat (4) Specific volume of steam. | 04 |
| (c) | A 4 cylinder, 4 stroke marine oil engine has a cylinder diameter of 490 mm and a piston stroke of 1000 mm. the engine uses 130 kg of fuel of calorific value 42000 KJ/kg in one hour when running at 2 rev/sec. the torque transmitted at the engine coupling is 22 KN.m and indicated mean effective pressure 710 KN/m ² . Determine (1) Indicated power (2) Brake power (3) Brake thermal efficiency (4) Mechanical efficiency (5) Indicated thermal efficiency. | 07 |

OR

- | | | |
|-----|--|-----------|
| (c) | A two-cylinder four stroke petrol engine has swept volume of $1.1 \times 10^{-3} \text{ m}^3$. It run at 950 rpm and consume 2.2 kg of petrol per hour having calorific value of 43000 KJ/kg. The mean effective pressure in both cylinder is 7.5 bar. Determine indicated thermal efficiency if clearance volume is 15% of swept volume. | 07 |
|-----|--|-----------|

- Q.3**
- | | | |
|-----|---|-----------|
| (a) | Explain physical properties of engineering materials. | 03 |
| (b) | Efficiency of Carnot cycle is independent of working fluid justify. | 04 |
| (c) | The initial volume of 0.9 kg of a certain gas was 0.75 m ³ at a temperature of 15 ⁰ C and a pressure of 1 bar. After adiabatic compression the volume is reduce to 0.28 m ³ and pressure was found to be 4 bar. Calculate (1) gas constant (2) molecular mass if $R_0 = 8314.3 \text{ J/kg mol K}$, (3) ratio of specific heats (4) C_p and C_v (5) change in internal energy | 07 |

OR

- Q.3**
- | | | |
|-----|---|-----------|
| (a) | What is thermal prime mover? Why they are most important prime movers? | 03 |
| (b) | What are nonferrous metals? Name any five and state their application. | 04 |
| (c) | Explain with neat sketch throttling calorimeter. Also state its advantages and disadvantages. | 07 |

- Q.4**
- | | | |
|-----|---|-----------|
| (a) | What is clutch? State its functions. | 03 |
| (b) | What is refrigerant? Describe the properties of good refrigerant. | 04 |

- (c) A cylindrical vessel of 1 m diameter and 4 m length has hydrogen gas at pressure of 100 KPa and 27⁰ C. Determine the amount of heat to be supplied so as to increase gas pressure to 125 KPa. For hydrogen take $C_p = 14.307 \text{ KJ/Kg K}$, $C_v = 10.183 \text{ KJ/Kg K}$. **07**

OR

- Q.4** (a) What is the difference between rigid coupling and flexible coupling? **03**
 (b) What is priming? Why priming is required in centrifugal pump but not in reciprocating pump? **04**
 (c) Air is to be compressed in a single acting reciprocating compressor from 1.013 bar and 150 C to 7 bar. Calculate the indicated power required for free air delivery of 0.3 m³/min. when the compression process is 1. Isentropic 2. Reversible isothermal 3. polytropic with $n = 1.25$. **07**

- Q.5** (a) Explain 1 ton of refrigeration and refrigeration effect. **03**
 (b) What are the advantages and disadvantages of water tube boiler over fire tube boiler? **04**
 (c) Explain construction and working of Babcock and Wilcox boiler with line diagram **07**

OR

- Q.5** (a) Economizer used to increase efficiency of boiler justify this statement. **03**
 (b) What are the differences between reciprocating and rotary compressor? **04**
 (c) Determine the efficiency of air standard Carnot cycle with the following data **07**

Minimum temperature of the cycle = 27⁰ C

Minimum pressure in the cycle = 1 bar

Pressure after isothermal compression = 4.5 bar

Pressure after isentropic compression = 12 bar

Take $R = 0.287 \text{ KJ/kg K}$.

Determine also power produced if engine makes 3 cycle/sec.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110006****Date:03-03-2023****Subject Name:Basic Mechanical Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Steam Table is allowed

- Q-1** (a) Define Specific heat and enthalpy. **03**
(b) Explain different between renewable and non-renewable energy. **04**
(c) Define adiabatic process. Prove that $pV^\gamma = \text{constant}$ **07**

- Q-2** (a) Explain classification of engineering materials. **03**
(b) Explain with neat sketch the working of fast and loose pulley. **04**
(c) How much heat to be added to convert 4kg water at 20⁰ C into steam at 8bar and 200⁰ C. take C_p of superheated steam as 2.1KJ/kg and specific heat of water of water as 4.187KJ/kgK. **07**

OR

- (c) 1kg of air at 7bar pressure and 90⁰C temperature undergoes a non flow polytropic process. The law of expansion is $pV^{1.1} = \text{constant}$. The pressure falls to 1.4bar during process. Calculate: (1) final temperature (2) Work done (3) change in internal energy (4) heat exchange take $R = 287\text{J/kg K}$ and $\gamma = 1.4$ for air. **07**

- Q-3** (a) Explain with neat sketch throttling calorimeter. **03**
(b) Explain different between fire tube boiler and water tube boiler. **04**
(c) Derive equation for efficiency of constant pressure heat addition cycle. **07**

OR

- Q-3** (a) Explain dryness fraction of steam. **03**
(b) The engine working on ideal Otto cycle. The temperature at the beginning and at the end of compression is 50⁰ C and 400⁰ C. Calculate the air standard efficiency and compression ratio. **04**
(c) Explain with neat sketch Cochran boiler. **07**

- Q-4** (a) Draw net figures of any three types of belt drive. **03**
(b) What are the uses of compressed air? **04**
(c) The following reading were taken during the test on a single cylinder four stroke oil engine : **07**
Cylinder diameter = 270 mm
Stroke length = 380 mm
Mean effective pressure = 6 bar
Engine speed = 250 rpm
Net load on brake = 1000 N
Effective mean diameter of brake = 1.5 m
Fuel used = 10 kg/hr
C.V of fuel = 44400KJ/kg
Calculate: (1) Brake power

- (2) Indicated power
- (3) Indicated thermal efficiency.

OR

- | | | | |
|------------|-----|--|-----------|
| Q-4 | (a) | Write function of clutch, Break and Coupling. | 03 |
| | (b) | Classify internal combustion Engine. | 04 |
| | (c) | Two cylinder four stroke diesel has total swept volume of 870cm ³ following data available with test on the engine.
Engine speed = 300rpm
Brake torque= 50 N-m
p _m = 10bar
calculate (1) indicated power
(2) brake power
(3) Mechanical efficiency | 07 |
| OR | | | |
| Q-5 | (a) | Explain difference between belt drive and chain drive | 03 |
| | (b) | Explain with the help of neat sketches, the working of two stroke petrol engine. | 04 |
| | (c) | Explain vapour compression refrigeration cycle. | 07 |
| OR | | | |
| Q-5 | (a) | Define the following mechanical properties
(i) Ductility (ii) Hardness (iii) Plasticity | 03 |
| | (b) | What is coupling? Explain internal expanding shoe brake with a neat sketch? | 04 |
| | (c) | What is priming? Explain with neat sketch centrifugal pump. | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023****Subject Code:3110006****Date:28-07-2023****Subject Name:Basic Mechanical Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Use of Steam Tables is permitted.

- Q.1** (a) Classify boilers according to (i) relative position of hot gases and water, (ii) axis of the shell, (iii) method of water circulation. Give examples in each. **03**
- (b) Make a list of (i) natural fuels and (ii) artificial (prepared) fuels. Write two effective points to reduce global warming. **04**
- (c) i. Determine the amount of heat required to raise the temperature of a steel workpiece from 40 to 160 °C. The mass of workpiece is 20 kg and specific heat is 460 J/kg K. **07**
- ii. A thermodynamic cycle has four processes. The heat and work interaction with surroundings is given in the table. Determine the work done during process 3-4. What will be the change in internal energy after completion of one cycle?

Process	Heat transfer (kJ)	Work done (kJ)
1-2	+925	+70
2-3	-110	-50
3-4	-770	?
4-1	+220	+170

- Q.2** (a) State the function of the following in boilers: feed check valve, blow off cock, fusible plug. **03**
- (b) Identify the following properties of materials: **04**
- 1) This property is desirable for parts subjected to impact loads and vibrations.
 - 2) By this property gold, silver, can be flattened or bent without cracking when hammered.
 - 3) This property is useful for materials subjected to high temperatures like boilers and turbines.
 - 4) This property is necessary for a material to be used in making springs.
- (c) Air has a volume of 0.15 m³, pressure 1.5 bar and temperature 107 °C. It is compressed at constant pressure, until its volume becomes 0.11 m³. Determine: (i) temperature at the end of compression, (ii) work done during compression, (iii) change in internal energy. $c_p = 1.005$ kJ/kg K, $c_v = 0.718$ kJ/kg K **07**

OR

- (c) 0.35 m³ of gas at 10 bar, 157 °C expands adiabatically to 4 bar. It is then compressed isothermally to its original volume. Find the final temperature and pressure of the gas. $c_p = 0.996$ kJ/kg K, $c_v = 0.703$ kJ/kg K **07**
- Q.3** (a) State the need of couplings along with the purposes they serve in machineries. **03**
- (b) Draw a schematic diagram of vapour compression refrigeration system. State the function of each main component. **04**

- (c) Steam is initially dry saturated at 9 bar in each of the following cases. 07
Determine,
1) its dryness fraction if it loses 50 kJ/kg heat at constant pressure,
2) the degree of superheat if it receives 150 kJ/kg heat at constant pressure.

OR

- Q.3 (a) Draw a half-sectional view of the sleeve (muff) coupling. 03
(b) Draw a labeled diagram of split air-conditioner. Why does the split air-conditioner consume more power than a window air-conditioner of same capacity? 04
(c) Draw a neat self-explanatory diagram of separating calorimeter. During the test on a separating calorimeter, 2.2 kg moisture was collected and 16 kg of steam left the calorimeter. Calculate dryness fraction of steam. 07

- Q.4 (a) Classify clutches. 03
(b) Draw a schematic diagram of reciprocating pump. Show the variation of discharge of water with crank angle for single-acting reciprocating pump. 04
(c) A Carnot cycle works with isentropic compression ratio of 6 and isothermal expansion ratio of 2. The volume of air at the beginning of isothermal expansion is 0.2 m^3 . If $T_{\max}$ and $p_{\max}$ is limited to 600 K and 20 bar respectively, determine: (i) minimum pressure during the cycle, (ii) thermal efficiency of cycle. Show the T - s and p - v diagram. 07

OR

- Q.4 (a) State the function of brakes and classify them. 03
(b) Draw a labeled figure of centrifugal pump indicating all main components. 04
(c) The compression ratio of an ideal air standard diesel cycle is 15. Heat supplied at constant pressure is 1470 kJ/kg of air. Show the cycle on p - v diagram and determine the cycle efficiency if inlet conditions are 300 K and 1 bar. 07

- Q.5 (a) Enlist various belt drives. Name any three belt materials. 03
(b) Outline the technical meaning of following terms used in air compressor: single acting compressor, single stage compressor, pressure ratio, swept volume, volumetric efficiency. 04
(c) Indicated power of a 6-cylinder 4-stroke engine is 150 kW at an average piston speed of 300 m/min. Stroke to bore ratio is 1.25. If mean effective pressure is 650 kN/m^2 , determine crankshaft speed. 07
What is a flywheel and what is its function in IC engine?

OR

- Q.5 (a) Match the following: 03

Spiral gears	Parallel shafts
Bevel gears	Intersecting shafts
Worm gears	Non-parallel non-intersecting shafts
Helical gears	Shaft axes at right angles & non-intersecting

- (b) Define with regard to compressors: mean effective pressure, indicated power, brake power, mechanical efficiency. 04
(c) A 4-cylinder 4-stroke petrol engine has a bore 60 mm and a stroke of 90 mm. The rated speed is 2800 rpm and torque is 55 N-m. Fuel consumption is 6.75 litre/hr. Specific gravity of petrol is 0.76 and calorific value is 44200 kJ/kg. Calculate brake power, brake mean effective pressure, brake thermal efficiency and brake specific fuel consumption. 07

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023****Subject Code:3110006****Date:29-01-2024****Subject Name:Basic Mechanical Engineering****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Use of Steam Tables is permitted.

Q.1 (a) What do you mean by 'boiler mountings' and 'boiler accessories'? Write three examples of them. **03**

(b) Explain advantages and disadvantages of wind energy. **04**

(c) i. Fill in the blanks: 1 bar = _____ Pa = _____ kPa = _____ MPa **07**

1 atm = _____ bar

ii. Write the equation of 1st law of thermodynamics for a stationary closed system.

iii. Determine the power required to accelerate a 900-kg car from rest to a velocity of 80 km/h in 20 sec on a horizontal road. (Neglect friction, rolling resistance, air drag, etc.)

Q.2 (a) State the function of the following in boilers: **03**
Economizer, air preheater, superheater.

(b) Provide meaning of the following statements regarding cast iron material: **04**

1) It is brittle and has low ductility.

2) It has high compressive strength but low tensile strength.

3) It has excellent castability.

4) It has poor formability.

(c) A gas whose pressure, volume, and temperature are 2.75 bar, 0.09 m³, and 185°C respectively has the state changed at constant pressure until its temperature becomes 15°C. Calculate (i) the heat transferred, and (ii) the work done. Take $R = 0.29$ kJ/kg K and $c_p = 1.005$ kJ/kg K. **07**

OR

(c) The gauge pressure of an automobile tire is measured to be 210 kPa before a trip and 220 kPa after the trip at a location where the atmospheric pressure is 95 kPa. Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine air temperature in the tire after the trip. Convert this change in temperature into percentage change. **07**

Q.3 (a) Classify clutches. **03**

(b) Draw a schematic diagram of the vapor absorption refrigeration system. **04**

(c) Determine the missing properties and the phase descriptions in the following table for water. Use steam tables and write necessary explanations. **07**

	$T, ^\circ\text{C}$	P, kPa	$h, \text{kJ/kg}$	x	Phase description
(i)		200		0.6	
(ii)	124		1835.2		

OR

- Q.3** (a) Draw a labeled diagram of a shoe brake used in automobiles. **03**
(b) Draw a self-explanatory diagram of window room air-conditioner. **04**
(c) Draw a neat, labeled diagram of barrel calorimeter. Write the equation used for this calorimeter for finding dryness fraction and state what each term in it indicates. **07**

- Q.4** (a) Match the following: **03**

Rigid flange coupling	Axis of shafts are parallel but not in alignment
Pin type flexible coupling	Where angular misalignment is more
Universal coupling	Requires proper alignment of shaft axes
Oldham's coupling	For small angular misalignment

- (b) Compare centrifugal pumps v/s reciprocation pumps. **04**
(c) Consider a steam power plant operating on the simple ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350°C and is condensed in the condenser at a pressure of 74 kPa. Draw the T - s diagram and determine the thermal efficiency of this cycle. (Do not neglect the pump work.) **07**

OR

- Q.4** (a) Give the similarities and differences between a clutch and a brake. **03**
(b) Explain any one type of rotary pumps with figure. **04**
(c) Derive, using P - v diagram, the equation to find air standard efficiency of Otto cycle. Use the formula you derived to find which gas from the following will give maximum efficiency for the same compression ratio: exhaust gas ($\gamma=1.3$), air, and monatomic gas ($\gamma=1.67$) **07**

- Q.5** (a) Draw a schematic diagram of the fast and loose pulley drive. **03**
(b) Give classification of air compressors. **04**
(c) A 4-stroke 6-cylinder IC engine has a stroke volume of 1.75 liters and is operating at a mean effective pressure of 6 bar. At what crankshaft rpm will the engine develop 35 hp? (Take 1 hp = 736 watts) **07**
What is the function of a carburetor and a fuel injector in an IC engine?

OR

- Q.5** (a) Differentiate between a belt drive and a gear drive. **03**
(b) Define with regard to compressors: double-acting compressor, multistage compressor, displacement (swept) volume, pressure ratio. **04**
(c) A petrol engine has a compression ratio of 6 and develops 15 kW. Its brake thermal efficiency is half of its air standard efficiency. Find the fuel consumption. Take calorific value of petrol as 41500 kJ/kg. **07**
Why are more two-wheeler automobiles made with a 4-stroke engine instead of a 2-stroke engine? Give three technical reasons for this.

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110006

Date:08-07-2024

Subject Name:Basic Mechanical Engineering

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Define these terms: (1) Absolute Pressure (2) Thermal Equilibrium (3) Prime Mover	03
(b) Explain with suitable examples the following thermodynamic systems: (1) Isolated (2) Open	04
(c) Write a short note on “Renewable Energy Sources”.	07
Q.2 (a) Define: (1) Isothermal process (2) Adiabatic process (3) Degree of superheat.	03
(b) Determine the condition of steam at a pressure of 10 bar and temperature of 200 °C.	04
(c) A gas initially at 510 kPa and volume of 142 liters undergoes a certain process after which the pressure and volume reach to 170 kPa and 275 liters respectively. And there is enthalpy drop of 65 kJ. The specific heat at constant volume for this gas is 718 J/kg-K. Calculate:1. Change in internal energy 2. Specific heat at constant pressure 3. Gas Constant.	07
OR	
(c) (1) Draw neat diagram of the Separating Calorimeter. (2) Determine the dryness fraction of steam after throttling from pressure 14 bar to 1.1 bar. The initial dryness fraction of steam is equal to 0.70	07
Q.3 (a) Define the terms: (1) T.D.C. (2) B.D.C. (3) Compression Ratio	03
(b) Classify the I. C. Engines.	04
(c) Calculate the air standard efficiency of the diesel cycle where the pressure and temperature at the beginning of compression process are 1 bar and 27 °C respectively. The compression ratio is 14 and the constant pressure heat addition continues up to 7% of the stroke volume. Draw this Diesel cycle on p-v diagram.	07

OR

- Q.3** (a) Write the function of following I.C. Engine components: **03**
(1) Spark Plug (2) Injector (3) Piston Rings
- (b) Calculate the air standard efficiency of an Otto Cycle engine having a bore of 100 mm and stroke length equal to 150 mm. The clearance volume of cylinder is equal to 250 cm^3 . Take $\gamma = 1.4$. **04**
- (c) A Diesel engine develops 5 kW brake power at a speed of 1000 rpm. **07**
The mechanical efficiency of the engine is 75% and indicated thermal efficiency is equal to 30 %. The diesel fuel having specific gravity = 0.87 and calorific value = 42700 kJ/kg is used.
Calculate: 1. Fuel consumption in kg/h. 2. Brake thermal efficiency 3. Brake specific fuel consumption.
- Q.4** (a) Define these terms: (1) Refrigerant (2) C.O.P. (3) Air conditioning. **03**
(b) Classify the Boilers. **04**
(c) Explain with neat diagrams the working of centrifugal compressor and centrifugal pump. **07**

OR

- Q.4** (a) What is an economizer? Why is it used in boiler plant? **03**
(b) Classify the Pumps. **04**
(c) Explain the characteristics of a Lancashire boiler with neat diagrams. **07**
- Q.5** (a) List the important properties of engineering materials. **03**
(b) Explain with neat diagram the working of domestic refrigerator. **04**
(c) Classify: (1) Clutches (2) Brakes **07**

OR

- Q.5** (a) Define these terms: (1) Coupling (2) Shaft (3) Spindle **03**
(b) Explain with neat diagram the construction and working of Roots Blower. **04**
(c) Compare: Belt, Chain and Gear drives. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:3110006

Date:02-01-2025

Subject Name:Basic Mechanical Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Simple and non-programmable scientific calculators are allowed.
6. Use of Steam Tables is permitted.

	Marks
Q.1 (a) State and explain zeroth law of thermodynamics.	03
(b) List sources of renewable energy.	04
(c) Define prime mover and classify it.	07
Q.2 (a) Derive characteristic equation of a perfect gas.	03
(b) Define following terms:	04
(i) Dryness fraction	
(ii) Degree of superheat	
(iii) Enthalpy of evaporation	
(iv) Enthalpy of superheated steam	
(c) One kg of gas is compressed polytropically from 150 kPa pressure and 290 K temperature to 750 kPa. The compression is according to law $pV^{1.3} = C$. find :	07
(i) Final temperature	
(ii) Work-done	
(iii) Change in internal energy	
(iv) Amount of heat transferred	
(v) Change in enthalpy,	
Take $R=0.287$ kJ/kg K and $c_p= 1.001$ kJ/kg K	
OR	
(c) Determine dryness fraction of steam supplied to a separating and throttling calorimeter.	07
Water separated in separating calorimeter = 0.45 kg	
Steam discharge from throttling calorimeter = 7 kg	
Steam pressure in main pipe = 1.2 MPa	
Barometer reading= 760 mm of Hg	
Manometer reading = 180 mm of Hg	
Temperature of steam after throttling = 140° C	
Take $C_{ps}=2.1$ kJ/kgK	
Q.3 (a) State limitations of Carnot gas power cycle.	03
(b) Compare 2- stroke engine with 4- stroke engine.	04
(c) Derive an expression of air standard efficiency of Otto cycle. Draw its P-V and T-S diagram.	07
OR	
Q.3 (a) State assumptions considered for analysis of air standard cycle.	03
(b) Derive an expression for thermal efficiency of rankine cycle.	04
(c) The following data is available for 2-stroke diesel engine:	07

Bore = 10 cm, stroke = 15 cm, engine speed=1000 rpm, torque developed= 58 N-m, $\eta_m = 80\%$, $\eta_{i,th} = 40\%$, calorific value of fuel =44000 kJ/kg.

Find: (i) Indicated power (ii) Mean effective pressure (iii) Brake specific fuel consumption.

- Q.4** (a) Define Compressor. State uses of compressed air. **03**
 (b) Discuss briefly location and function of the followings; **04**
 (i) Water level indicator
 (ii) Pressure Gauge
 (c) Explain with neat sketch working of Cochran boiler. **07**
OR
- Q.4** (a) Define following with reference to air compressor; **03**
 (i) Free air delivery
 (ii) Swept volume
 (iii) Volumetric efficiency
 (b) Differentiate between boiler mountings and accessories. **04**
 (c) Explain with neat sketch construction and working of split air conditioner. **07**
- Q.5** (a) State the application of following gears **03**
 (i) Helical gear, (ii) worm and worm gear (iii) rack and pinion
 (b) What is coupling? Classify it. **04**
 (c) Explain with neat sketch working of centrifugal pump. **07**
OR
- Q.5** (a) Enlist types of belt drive. **03**
 (b) Differentiate between clutch and brake. **04**
 (c) Explain following properties of materials. **07**
 (i) Elasticity, (ii) Plasticity, (iii) Ductility, (iv) Brittleness (v) Hardness (vi) Toughness, (vii) Fatigue

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:3110006

Date:05-07-2025

Subject Name:Basic Mechanical Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Usage of steam table is permitted.

		Marks
Q.1	(a) Write a short note on “Global Warming and solar energy”	03
	(b) Write the mathematical form of the First Law and explain each term.	04
	(c) Derive the characteristics gas equation for a perfect gas with usual notations.	07
Q.2	(a) Differentiate between heat and work in the context of the First Law.	03
	(b) Name the types of prime movers based on the source of energy.	04
	(c) Explain vapour compression refrigeration cycle used in domestic refrigerator.	07
	OR	
	(c) How much heat to be added to convert 4kg water at 20 °C into steam at 8 bar and 200 °C. take C_p of superheated steam as 2.1 KJ/kg and specific heat of water of water as 4.187 KJ/kgK.	07
Q.3	(a) With usual notations prove that $C_p - C_v = R$.	03
	(b) The engine working on ideal Otto cycle. The temperature at the beginning and at the end of compression is 50 °C and 400 °C. Calculate the air standard efficiency and compression ratio.	04
	(c) Explain with neat sketch throttling calorimeter. Also state its advantages and disadvantages.	07
	OR	
Q.3	(a) Explain open system, closed system and isolated system.	03
	(b) Write a difference between SI engine and CI engine	04
	(c) Derive an expression of air standard efficiency of Diesel cycle. Draw its P-V and T-S diagram.	07
Q.4	(a) Classify internal combustion Engine.	03
	(b) List out application of compressed air.	04
	(c) Explain with neat sketch Cochran boiler.	07
	OR	
Q.4	(a) How do you classify steam boilers?	03
	(b) Draw labeled diagram of Babcock and Wilcox boiler.	04
	(c) What is priming? Explain with neat sketch centrifugal pump.	07
Q.5	(a) Explain single acting reciprocating pump.	03
	(b) Write a short on a single plate (disc) friction clutch	04
	(c) Define following mechanical properties: (1) Elasticity (2) Malleability (3) Ductility (4) Stiffness (5) Hardness (6) Toughness (7) Resilience.	07

OR

- | | | | |
|------------|------------|---|-----------|
| Q.5 | (a) | Write function of clutch, Break and Coupling. | 03 |
| | (b) | What is refrigerant? Describe the properties of good refrigerant. | 04 |
| | (c) | Compare belt drive, chain drive and gear drive. | 07 |

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: 3110006

Date: 13-01-2026

Subject Name: Basic Mechanical Engineering

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of Steam Tables is permitted.

	Marks
Q.1 (a) Give classification of Cochran boiler and Lancashire boiler.	03
(b) Write a short note on CNG as a fuel.	04
(c) (i) The absolute pressure in a compressed air tank is 200 kPa. What is the gauge pressure in the tank if atmospheric pressure is 1.01 bar? (ii) The temperature of a system rises by 130°C during a heating process. Express this rise in temperature in kelvins. (iii) A 4-kW resistance heater in a water heater runs for 3 hours to raise the water temperature to the desired level. Determine the amount of electric energy used in both kWh and kJ.	07
Q.2 (a) State the function of any three mountings in boilers.	03
(b) Classify engineering materials.	04
(c) 0.4 kg of gas is expanded isentropically from 10 bar and 340°C to 1 bar. It is then heated at constant volume to 3 bar and 340 °C and then finally it is compressed isothermally until the initial pressure of 10 bar is attained. Draw the p-V diagram for these processes and find the value of the adiabatic index γ . Take $C_p = 1.005 \text{ kJ/kgK}$.	07

OR

(c) A gas expands from 450 kPa and 130 litres to 150 kPa and 260 litres. The decrease in enthalpy during the process is 55 kJ. Taking $C_v = 718 \text{ J/kgK}$, determine (i) change in internal energy, (ii) value of C_p , and (iii) value of R.	07
Q.3 (a) What is a rigid coupling? What are its types?	03
(b) Draw a schematic diagram of the vapor compression refrigeration system. Where is the condenser located in a split air-conditioner?	04
(c) A rigid tank contains 10 kg of water at 90°C. If 8 kg of the water is in the liquid form and the rest is in the vapor form, determine (a) the pressure in the tank and (b) the volume of the tank.	07

OR

(a) What is a friction clutch? What are its types?	03
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- (b) What is the difference between the working principle of vapor compression and vapor absorption refrigeration system? 04
- (c) Find the dryness fraction of steam supplied in a combined separating-and-throttling calorimeter from following data: initial pressure = 10 bar, final pressure = 1 bar, water separated = 1.5 kg, steam discharged from throttling calorimeter = 20 kg, temperature of steam after throttling = 120°C. Take specific heat of superheated steam as 2.1 kJ/kg K. 07

- Q.4** (a) Find the incorrect statements from the following and correct them. 03
- Rigid flange coupling is used when axes of shafts are parallel but not in alignment.
 - Pin type flexible coupling is used where angular misalignment is more.
 - Universal coupling requires proper alignment of shaft axes.
 - Oldham's coupling is for small angular misalignment.
- (b) Describe main parts of a centrifugal pump in short. 04
- (c) A Diesel engine has a compression ratio of 15 and heat addition at constant pressure takes place 6% of the stroke. Find the air standard efficiency of the engine. 07

OR

- (a) Draw a labeled diagram of a simple band brake. 03
- (b) Explain any one type of rotary pumps with figure. 04
- (c) In an air standard Otto cycle, the upper and lower limits of absolute temperatures are T_3 and T_1 respectively. Show that for maximum work, the ratio of compression should have the value $r = \left(\frac{T_3}{T_1}\right)^{1.25}$. 07

- Q.5** (a) Which mechanical drive will you suggest in each of the following different situations: 03
- When there is less space available.
 - Where 'slip' is permitted.
 - When high velocity ratio is required.
- (b) What is isothermal efficiency, clearance ratio, and volumetric efficiency of a compressor? 04
- (c) Distinguish between petrol engine and diesel engine. Use the following points: cycle of operation, compression ratio, fuel ignition, governing, engine speed, thermal efficiency, engine weight. 07

OR

- (a) State three disadvantages of using gear drives. 03
- (b) Give classification of air compressors. 04
- (c) The following data refer to a test on I.C. engine: 07
- Indicated power = 42 kW, frictional power = 7 kW, engine speed = 1800 rpm, specific fuel consumption per B.P. = 0.30 kg/kWh, calorific value of fuel used = 43000 kJ/kg. Calculate: (i) mechanical efficiency, (ii) brake thermal efficiency, and (iii) indicated thermal efficiency.
